less clearly defined in the services sector. As a result, while ITIs offer almost an equal number of courses in engineering and non-engineering trades (86 and 78, respectively), 85 per cent of students enrol in manufacturing trades, as a 2026 NITI Aayog working paper, 'Education and Skilling for Employment: From Credentials to Learning Outcomes', notes.

Trainers, who play a critical role in determining learning outcomes, are in short supply too. As of July 2025, only 46.9 per cent of the 217,724 sanctioned trainer positions had been filled across ITIs, according to the NITI Aayog working paper.



"ITI courses have to be determined based on the hyperlocal requirements of the industry in that area"

SABINA DEWAN
Founder & Executive Director, JustJobs Network

Privatisation of ITIs has added another layer of concern over their quality. According to 'State of Working India 2026' report, much of the expansion since the 2000s has been driven by the private sector. Over 70 per cent of ITIs are now privately run, compared with the pre-1980s era, when 80 per cent were government-run. An analysis of the MSDE's ITI rankings suggests that private ITIs are more likely to fall short on quality parameters—including instructor qualifications, pedagogy and enrolment—than government-run institutions.

Beyond infrastructure, trainer quality and curricula, the ITI skilling ecosystem, says Sabina Dewan, founder and executive director of JustJobs Network, needs to be hyperlocalised. Currently, skilling courses are uniform across the country when they should instead respond to local labour-market needs. "Courses have to be determined based on the hyperlocal requirements of the industry in that area," she says. She cites the example of Ambala, which once had a thriving ecosystem of medical device manufacturers, an industry that eventually died because, among other reasons, ITIs in the region didn't train talent for the sector. "When there is granular local labour-demand data and training is mapped to that demand, real change will happen," says Dewan.

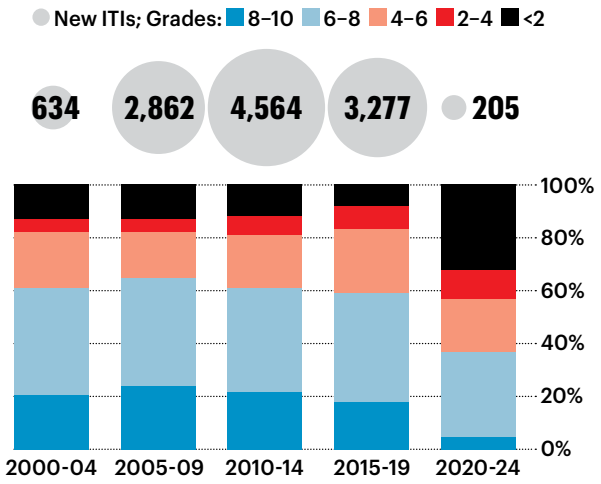
The government says it is moving towards such decentralisation. According to the MSDE, skill development plans are now being prepared at the state and district levels, mapping local industries, workforce availability, migration patterns, training capacity and emerging job roles. In 2024-25, 746 district plans and 28 state plans were submitted. The ministry is working to link scheme targets and public expenditure with district-level demand. "This represents an important shift—from allocating training opportunities administratively to planning them around the economic character of each region," the ministry says.

Projected shortages underscore the urgency. Staffing firm Quess estimates that, by 2031, India could face a shortage of 1.8 million workers in construction and infrastructure, 1.3 million skilled-trade professionals—including electricians, welders, plumbers, fitters and HVAC technicians—598,026 nurses and allied health providers, and 946,543 workers in tourism and hospitality. Across 19 sectors, the cumulative manpower shortage could reach close to 9.9 million.

Rahul Gama, CHRO, CEAT, says the roles hardest to fill are the ones at the intersection of traditional craft and new-age technology: skilled machine operators who can work with automated and semi-automated curing and moulding lines; maintenance technicians who understand both mechanical and electronic systems; and quality control inspectors trained in modern testing and analytics tools. "These roles demand a working knowledge of digital dashboards, basic troubleshooting logic, and safety protocol far beyond what traditional curricula cover," he says. The talent pipeline, he adds, has not kept pace with the rapid modernisation of shop floors. "Many applicants come in with theoretical certification but need three to six months of structured on-the-job training before they can become truly productive."

ITIs: A WANING PROMISE

Newer ITIs fare much worse on the quality score compared to ITIs incorporated in earlier years. For instance, only 6 per cent of the ITIs established in 2020-24 are in the top 8–10 grade; 32% are in the lowest (<2) grade



Source: Azim Premji University's report 'State of Working India 2026'; score of 8–10 indicates best quality, 6–8 good, 4–6 average, 2–4 below average, and below 2 poor. Numbers on the top of the bars refer to the number of ITIs incorporated during that period; % denotes share in final grading